

Logistic Regression

Logistic Regression is for **binary classification tasks**. It is mainly employed when the target variable is categorical with two possible classes or labels, typically represented as 0 and 1 or “Yes” and “No”.

For example

- 1) Fraud or Not
- 2) Email spam or not Spam
- 3) Infected with Disease or not Disease.

There are two types of approaches to compute logistic regression Graphical and Probabilistic approach.

Binary Class

```
from sklearn.linear_model import LogisticRegression  
  
model = LogisticRegression(solver='liblinear', max_iter=1000)
```

Multi Class

```
from sklearn.linear_model import LogisticRegression  
  
model = LogisticRegression(multi_class='ovr', solver='liblinear', max_iter=1000)
```

Initialize slope m and intercept c .

Calculate $z = mx + c$.

Apply the sigmoid function to get the predicted probability.

Calculate the error/loss.

Update m and c using gradient descent.

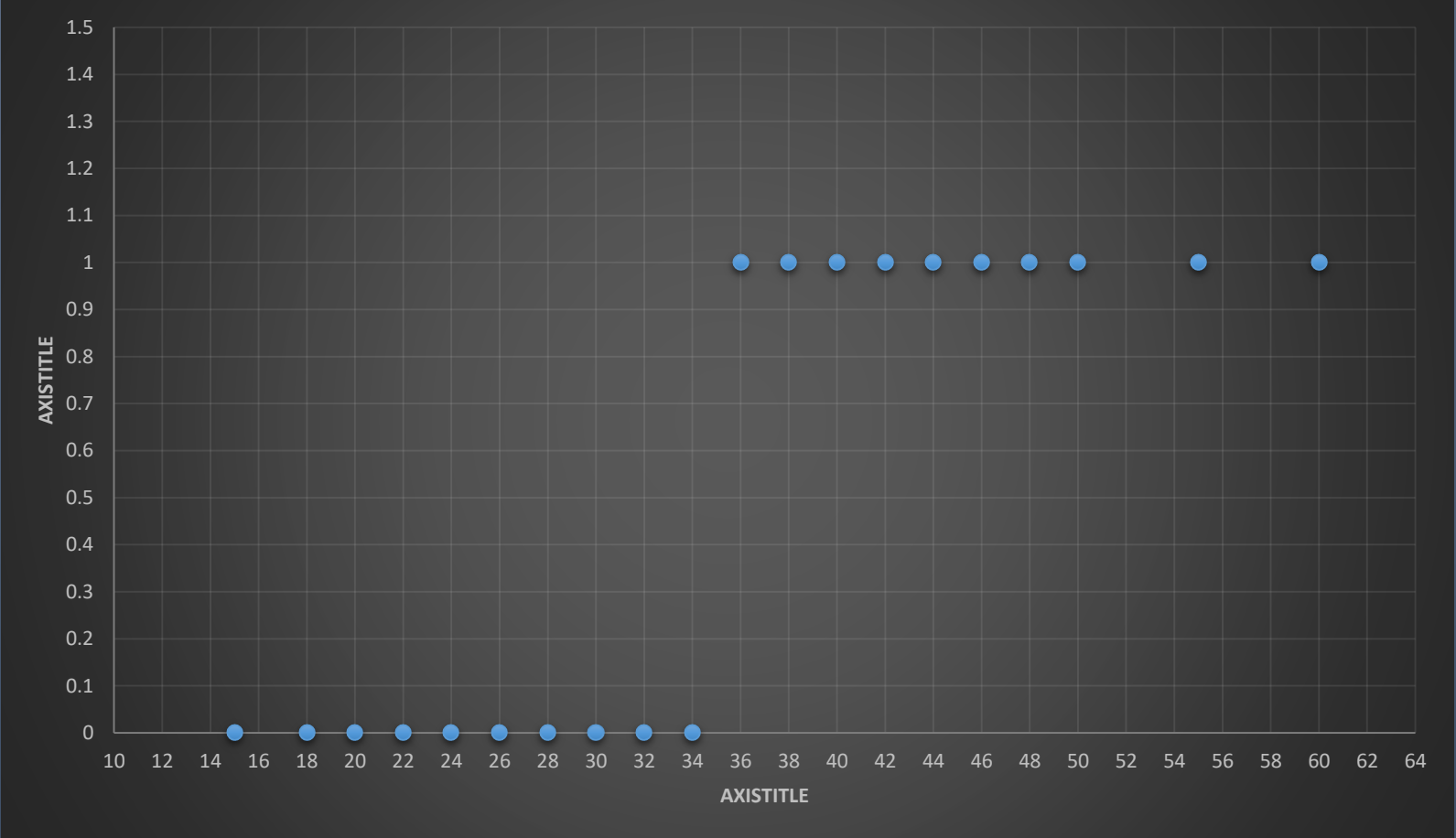
Repeat steps for multiple iterations.

Stop when the loss converges or the maximum iterations are reached.

Use the final m and c to make predictions.

Monthly Income (₹1000s)	ActualLoanApproved
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15	0
18	0
20	0
22	0
24	0
26	0
28	0
30	0
32	0
34	0
36	1
38	1
40	1
42	1
44	1
46	1
48	1
50	1
55	1
60	1

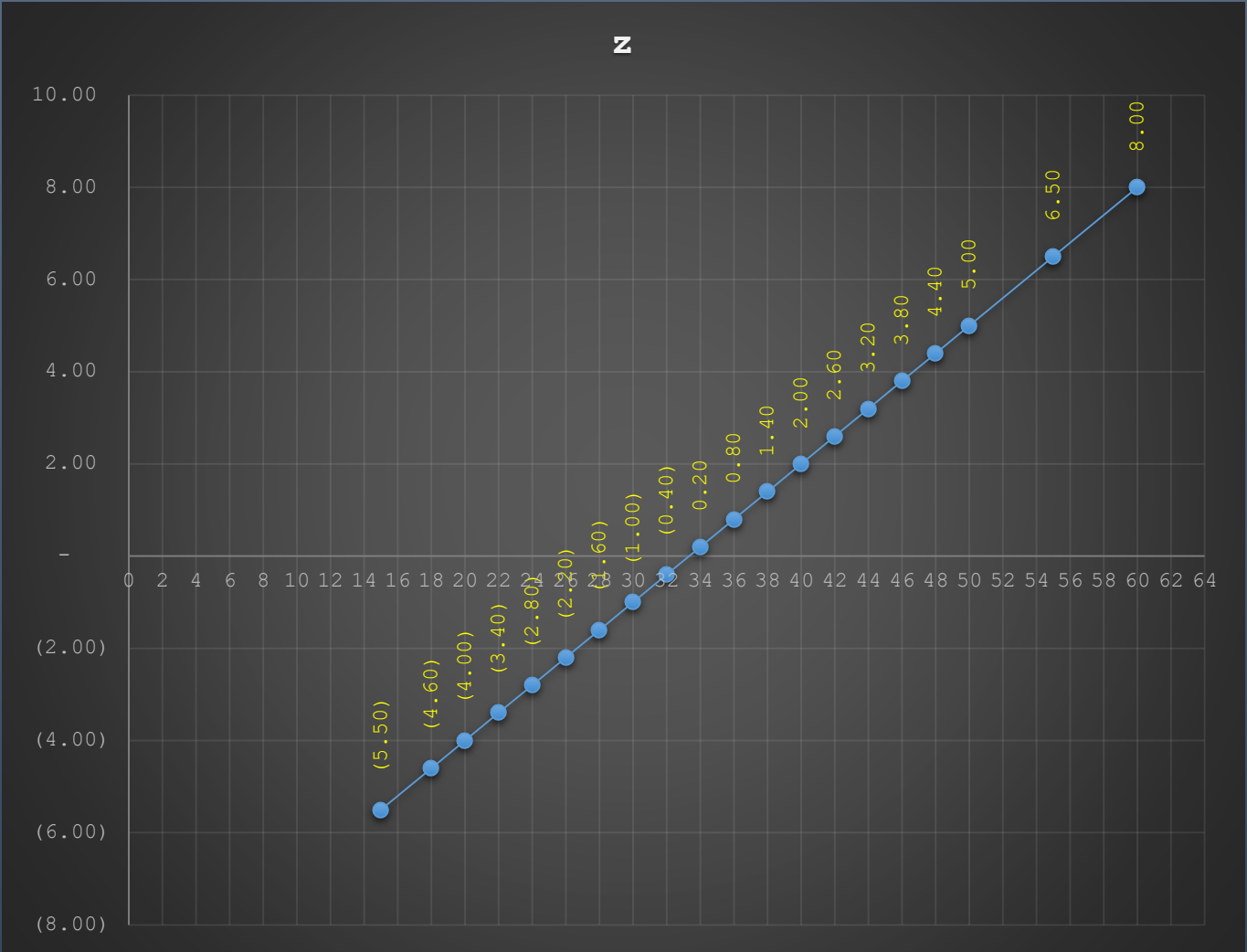


Intercept (c) = -10 Slope = 0.3

Monthly Income (₹1000s)	Actual Loan Approved	$mx+c$	z
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15	0	$= (0.3 \times 15) + -10$	-5.50
18	0	$= (0.3 \times 18) + -10$	-4.60
20	0	$= (0.3 \times 20) + -10$	-4.00
22	0	$= (0.3 \times 22) + -10$	-3.40
24	0	$= (0.3 \times 24) + -10$	-2.80
26	0	$= (0.3 \times 26) + -10$	-2.20
28	0	$= (0.3 \times 28) + -10$	-1.60
30	0	$= (0.3 \times 30) + -10$	-1.00
32	0	$= (0.3 \times 32) + -10$	-0.40
34	0	$= (0.3 \times 34) + -10$	0.20
36	1	$= (0.3 \times 36) + -10$	0.80
38	1	$= (0.3 \times 38) + -10$	1.40
40	1	$= (0.3 \times 40) + -10$	2.00
42	1	$= (0.3 \times 42) + -10$	2.60
44	1	$= (0.3 \times 44) + -10$	3.20
46	1	$= (0.3 \times 46) + -10$	3.80
48	1	$= (0.3 \times 48) + -10$	4.40
50	1	$= (0.3 \times 50) + -10$	5.00
55	1	$= (0.3 \times 55) + -10$	6.50
60	1	$= (0.3 \times 60) + -10$	8.00

$z = mx + c$



Monthly Income (₹1000s)	Actual Loan Approved	mx+c	z	After Sigmoid
15	0	= (0.3*15) + -10	-5.50	0.00
18	0	= (0.3*18) + -10	-4.60	0.01
20	0	= (0.3*20) + -10	-4.00	0.02
22	0	= (0.3*22) + -10	-3.40	0.03
24	0	= (0.3*24) + -10	-2.80	0.06
26	0	= (0.3*26) + -10	-2.20	0.10
28	0	= (0.3*28) + -10	-1.60	0.17
30	0	= (0.3*30) + -10	-1.00	0.27
32	0	= (0.3*32) + -10	-0.40	0.40
34	0	= (0.3*34) + -10	0.20	0.55
36	1	= (0.3*36) + -10	0.80	0.69
38	1	= (0.3*38) + -10	1.40	0.80
40	1	= (0.3*40) + -10	2.00	0.88
42	1	= (0.3*42) + -10	2.60	0.93
44	1	= (0.3*44) + -10	3.20	0.96
46	1	= (0.3*46) + -10	3.80	0.98
48	1	= (0.3*48) + -10	4.40	0.99
50	1	= (0.3*50) + -10	5.00	0.99
55	1	= (0.3*55) + -10	6.50	1.00
60	1	= (0.3*60) + -10	8.00	1.00

$$\text{SigmoidActivation} = \frac{1}{1 + e^{-z}}$$

$$e = 2.718$$

$$= \frac{1}{1 + 2.718^{-(-5.50)}}$$

$$= 0.00$$

$$= \frac{1}{1 + 2.718^{-(-4.60)}}$$

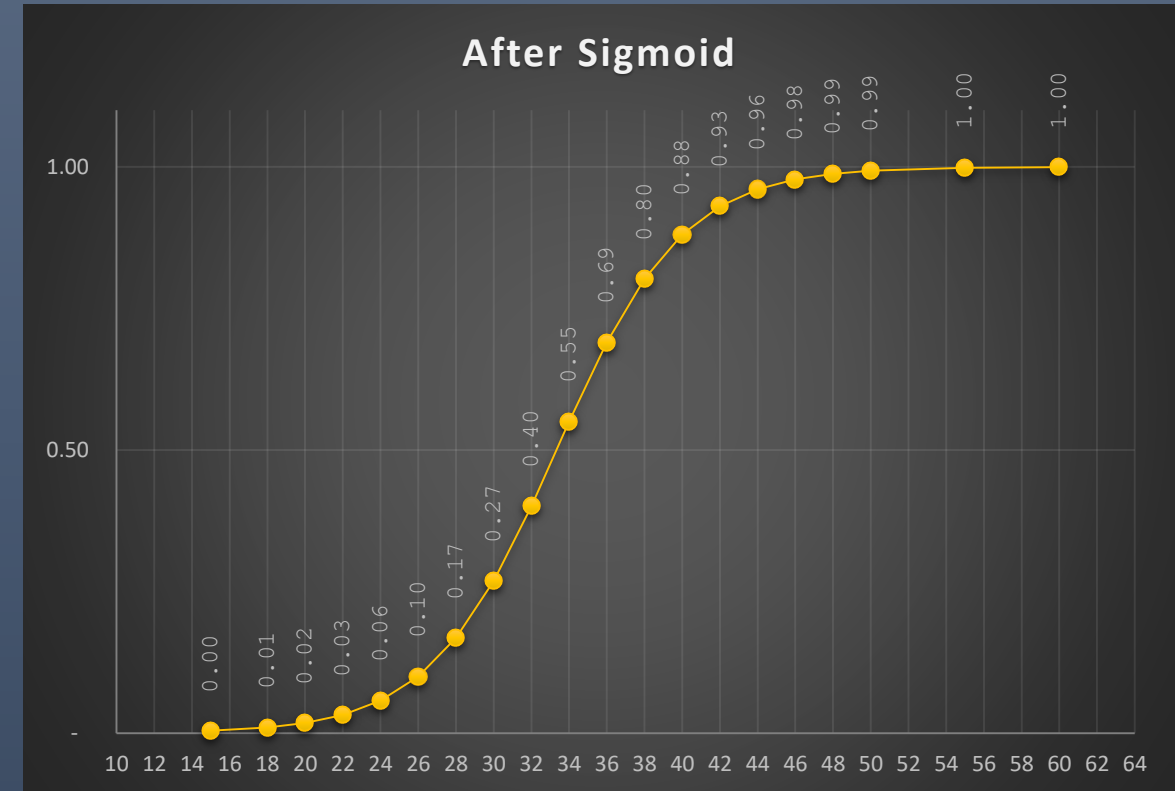
$$= 0.01$$

$$= \frac{1}{1 + 2.718^{-(-4.00)}}$$

$$= 0.02$$

Intercept (c) = -10 Slope = 0.3

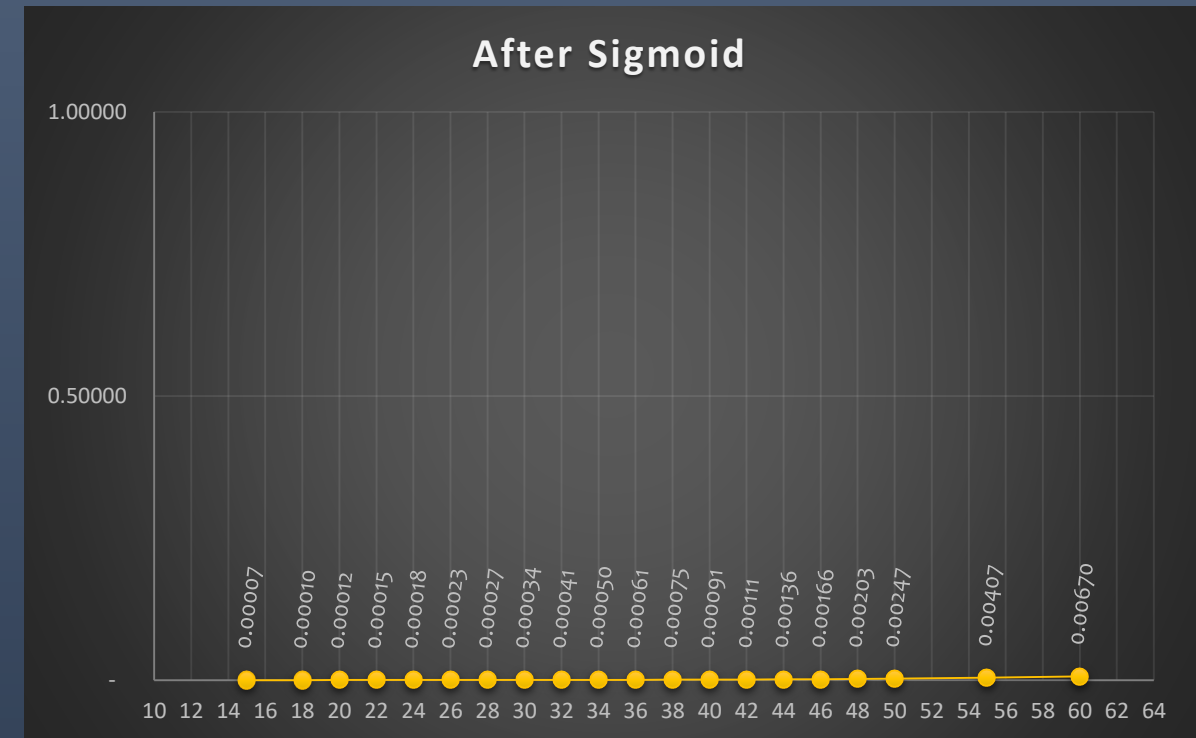
Monthly Income (₹1000s)	Actual Loan Approved	$mx+c$	z	After Sigmoid
15	0	$= (0.3 \times 15) + -10$	-5.50	0.00
18	0	$= (0.3 \times 18) + -10$	-4.60	0.01
20	0	$= (0.3 \times 20) + -10$	-4.00	0.02
22	0	$= (0.3 \times 22) + -10$	-3.40	0.03
24	0	$= (0.3 \times 24) + -10$	-2.80	0.06
26	0	$= (0.3 \times 26) + -10$	-2.20	0.10
28	0	$= (0.3 \times 28) + -10$	-1.60	0.17
30	0	$= (0.3 \times 30) + -10$	-1.00	0.27
32	0	$= (0.3 \times 32) + -10$	-0.40	0.40
34	0	$= (0.3 \times 34) + -10$	0.20	0.55
36	1	$= (0.3 \times 36) + -10$	0.80	0.69
38	1	$= (0.3 \times 38) + -10$	1.40	0.80
40	1	$= (0.3 \times 40) + -10$	2.00	0.88
42	1	$= (0.3 \times 42) + -10$	2.60	0.93
44	1	$= (0.3 \times 44) + -10$	3.20	0.96
46	1	$= (0.3 \times 46) + -10$	3.80	0.98
48	1	$= (0.3 \times 48) + -10$	4.40	0.99
50	1	$= (0.3 \times 50) + -10$	5.00	0.99
55	1	$= (0.3 \times 55) + -10$	6.50	1.00
60	1	$= (0.3 \times 60) + -10$	8.00	1.00



Monthly Income (₹1000s)	Actual Loan Approved	$mx+c$	$mx+c$	After Sigmoid	0 and 1	Final Correct
15	0	$= (0.3 * 15) + -10$	-5.50	0.00	0	Correct
18	0	$= (0.3 * 18) + -10$	-4.60	0.01	0	Correct
20	0	$= (0.3 * 20) + -10$	-4.00	0.02	0	Correct
22	0	$= (0.3 * 22) + -10$	-3.40	0.03	0	Correct
24	0	$= (0.3 * 24) + -10$	-2.80	0.06	0	Correct
26	0	$= (0.3 * 26) + -10$	-2.20	0.10	0	Correct
28	0	$= (0.3 * 28) + -10$	-1.60	0.17	0	Correct
30	0	$= (0.3 * 30) + -10$	-1.00	0.27	0	Correct
32	0	$= (0.3 * 32) + -10$	-0.40	0.40	0	Correct
34	0	$= (0.3 * 34) + -10$	0.20	0.55	1	-
36	1	$= (0.3 * 36) + -10$	0.80	0.69	1	Correct
38	1	$= (0.3 * 38) + -10$	1.40	0.80	1	Correct
40	1	$= (0.3 * 40) + -10$	2.00	0.88	1	Correct
42	1	$= (0.3 * 42) + -10$	2.60	0.93	1	Correct
44	1	$= (0.3 * 44) + -10$	3.20	0.96	1	Correct
46	1	$= (0.3 * 46) + -10$	3.80	0.98	1	Correct
48	1	$= (0.3 * 48) + -10$	4.40	0.99	1	Correct
50	1	$= (0.3 * 50) + -10$	5.00	0.99	1	Correct
55	1	$= (0.3 * 55) + -10$	6.50	1.00	1	Correct
60	1	$= (0.3 * 60) + -10$	8.00	1.00	1	Correct

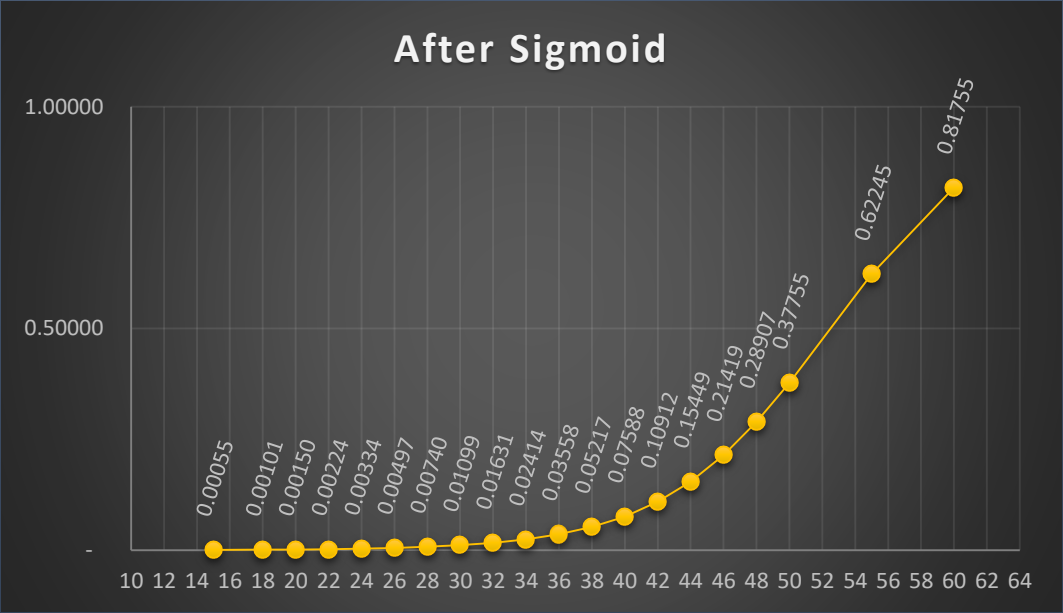
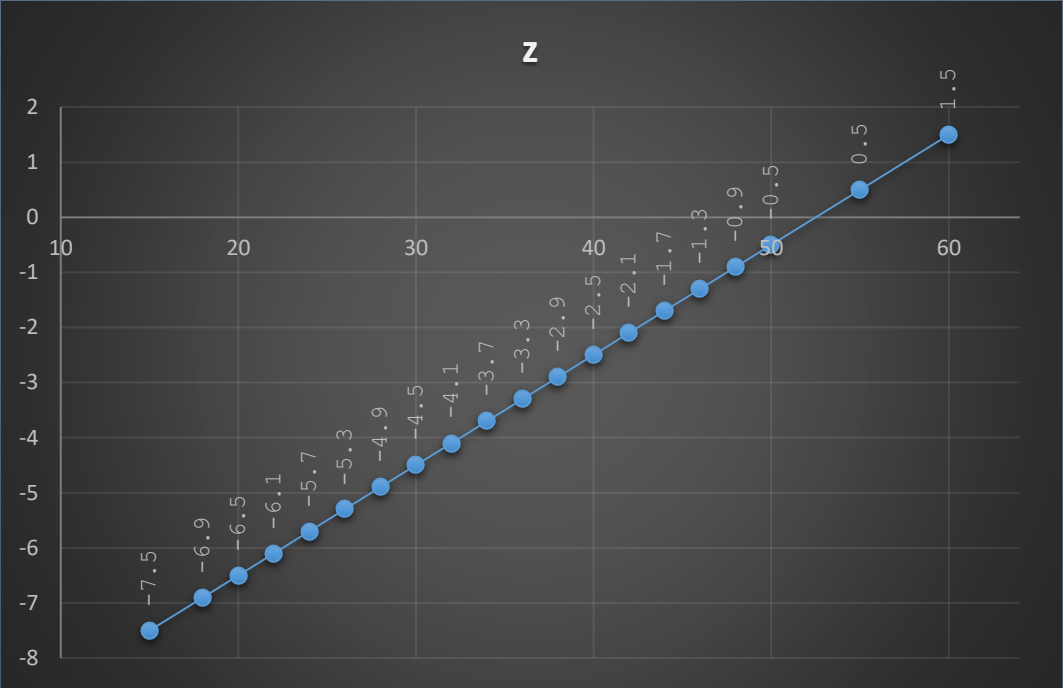
Intercept (c)= -11 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-9.5	0.00007	0	Correct
18	0	-9.2	0.00010	0	Correct
20	0	-9.0	0.00012	0	Correct
22	0	-8.8	0.00015	0	Correct
24	0	-8.6	0.00018	0	Correct
26	0	-8.4	0.00023	0	Correct
28	0	-8.2	0.00027	0	Correct
30	0	-8.0	0.00034	0	Correct
32	0	-7.8	0.00041	0	Correct
34	0	-7.6	0.00050	0	Correct
36	1	-7.4	0.00061	0	-
38	1	-7.2	0.00075	0	-
40	1	-7.0	0.00091	0	-
42	1	-6.8	0.00111	0	-
44	1	-6.6	0.00136	0	-
46	1	-6.4	0.00166	0	-
48	1	-6.2	0.00203	0	-
50	1	-6.0	0.00247	0	-
55	1	-5.5	0.00407	0	-
60	1	-5.0	0.00670	0	-



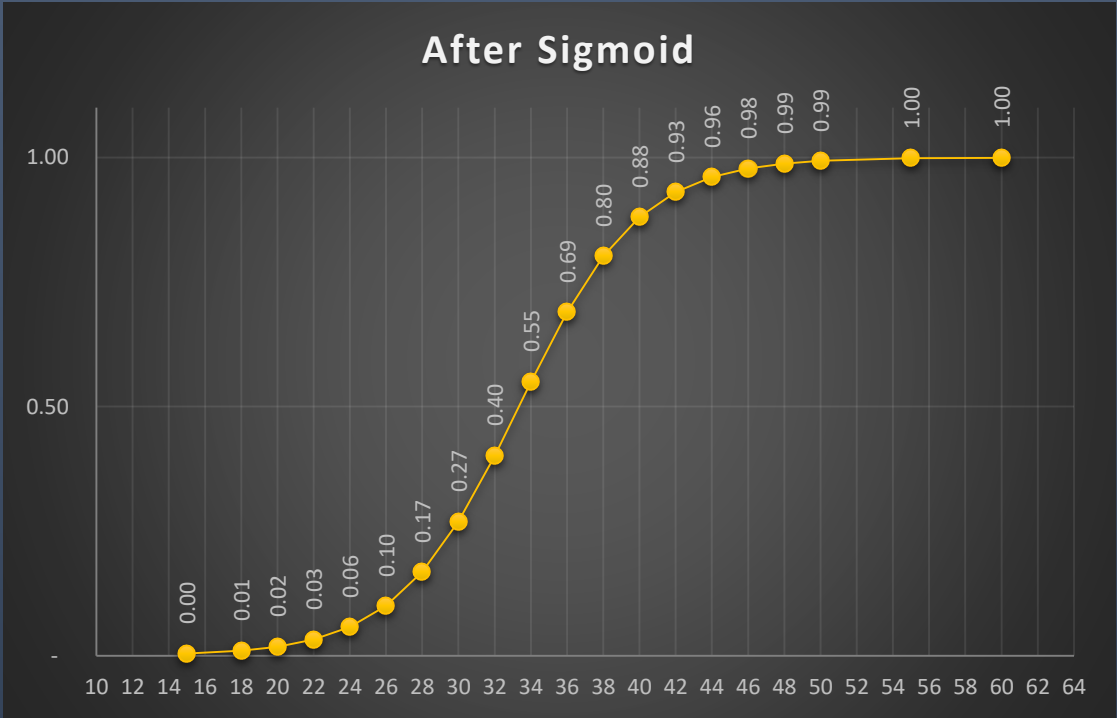
Intercept(c)= -10.5 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-7.5	0.00055	0	Correct
18	0	-6.9	0.00101	0	Correct
20	0	-6.5	0.00150	0	Correct
22	0	-6.1	0.00224	0	Correct
24	0	-5.7	0.00334	0	Correct
26	0	-5.3	0.00497	0	Correct
28	0	-4.9	0.00740	0	Correct
30	0	-4.5	0.01099	0	Correct
32	0	-4.1	0.01631	0	Correct
34	0	-3.7	0.02414	0	Correct
36	1	-3.3	0.03558	0	-
38	1	-2.9	0.05217	0	-
40	1	-2.5	0.07588	0	-
42	1	-2.1	0.10912	0	-
44	1	-1.7	0.15449	0	-
46	1	-1.3	0.21419	0	-
48	1	-0.9	0.28907	0	-
50	1	-0.5	0.37755	0	-
55	1	0.5	0.62245	1	Correct
60	1	1.5	0.81755	1	Correct



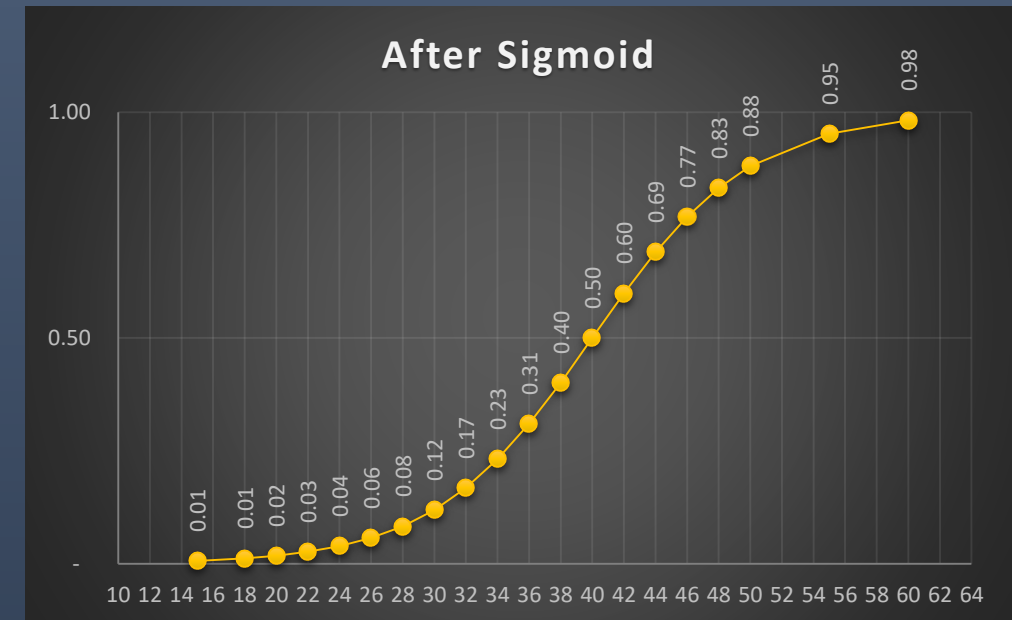
Intercept (c)= -10 Slope = 0.3

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.50	0.00	0	Correct
18	0	-4.60	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.40	0.03	0	Correct
24	0	-2.80	0.06	0	Correct
26	0	-2.20	0.10	0	Correct
28	0	-1.60	0.17	0	Correct
30	0	-1.00	0.27	0	Correct
32	0	-0.40	0.40	0	Correct
34	0	0.20	0.55	1	-
36	1	0.80	0.69	1	Correct
38	1	1.40	0.80	1	Correct
40	1	2.00	0.88	1	Correct
42	1	2.60	0.93	1	Correct
44	1	3.20	0.96	1	Correct
46	1	3.80	0.98	1	Correct
48	1	4.40	0.99	1	Correct
50	1	5.00	0.99	1	Correct
55	1	6.50	1.00	1	Correct
60	1	8.00	1.00	1	Correct



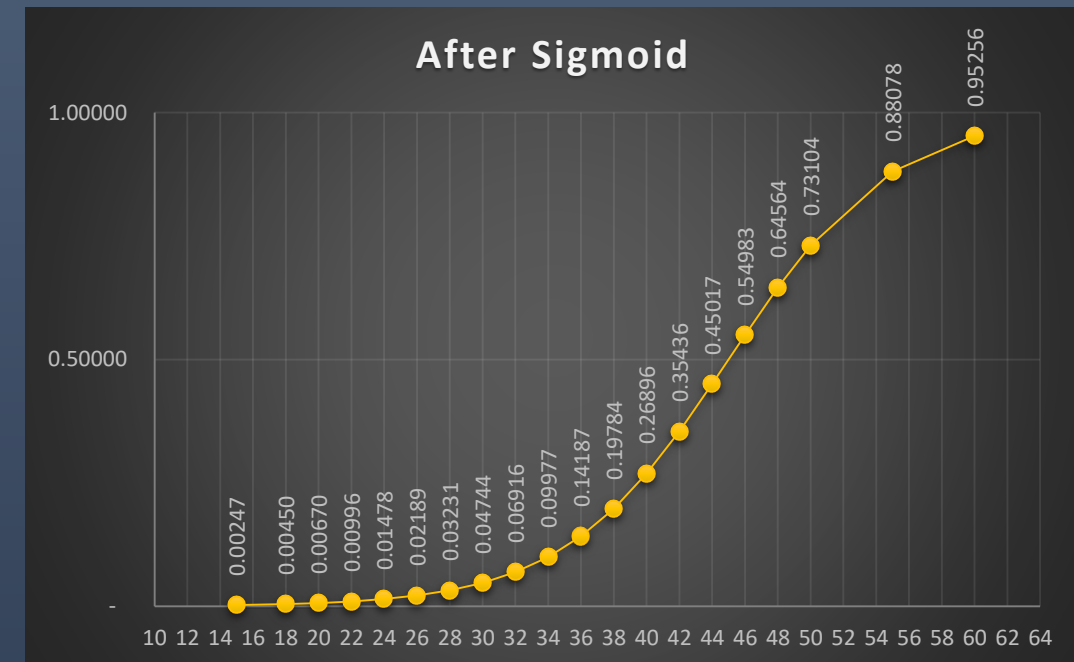
Intercept (c)= -8 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.00	0.01	0	Correct
18	0	-4.40	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.60	0.03	0	Correct
24	0	-3.20	0.04	0	Correct
26	0	-2.80	0.06	0	Correct
28	0	-2.40	0.08	0	Correct
30	0	-2.00	0.12	0	Correct
32	0	-1.60	0.17	0	Correct
34	0	-1.20	0.23	0	Correct
36	1	-0.80	0.31	0	-
38	1	-0.40	0.40	0	-
40	1	-	0.50	1	Correct
42	1	0.40	0.60	1	Correct
44	1	0.80	0.69	1	Correct
46	1	1.20	0.77	1	Correct
48	1	1.60	0.83	1	Correct
50	1	2.00	0.88	1	Correct
55	1	3.00	0.95	1	Correct
60	1	4.00	0.98	1	Correct

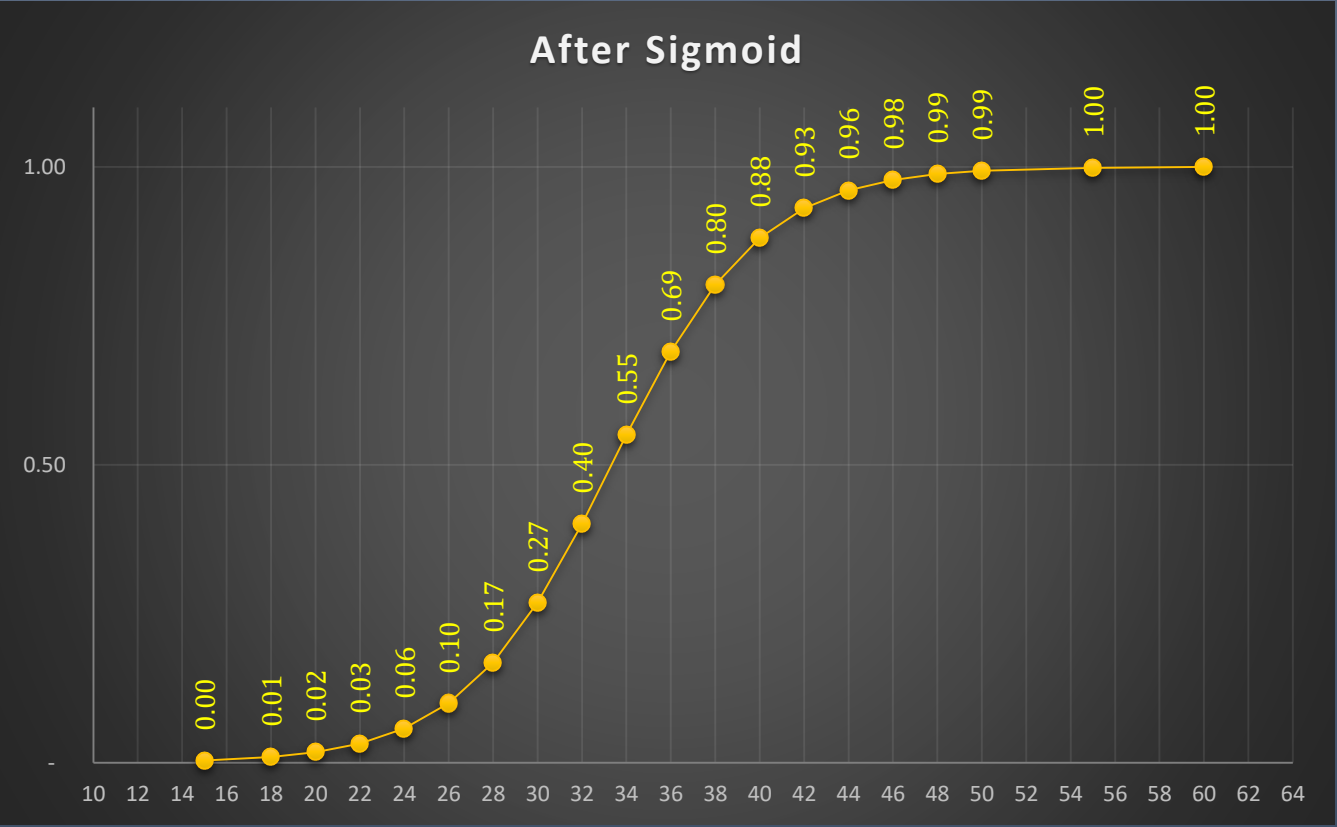


Intercept (c)= -9 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0 and 1	Final Correct
15	0	-6.00	0.00247	0	Correct
18	0	-5.40	0.00450	0	Correct
20	0	-5.00	0.00670	0	Correct
22	0	-4.60	0.00996	0	Correct
24	0	-4.20	0.01478	0	Correct
26	0	-3.80	0.02189	0	Correct
28	0	-3.40	0.03231	0	Correct
30	0	-3.00	0.04744	0	Correct
32	0	-2.60	0.06916	0	Correct
34	0	-2.20	0.09977	0	Correct
36	1	-1.80	0.14187	0	-
38	1	-1.40	0.19784	0	-
40	1	-1.00	0.26896	0	-
42	1	-0.60	0.35436	0	-
44	1	-0.20	0.45017	0	-
46	1	0.20	0.54983	1	Correct
48	1	0.60	0.64564	1	Correct
50	1	1.00	0.73104	1	Correct
55	1	2.00	0.88078	1	Correct
60	1	3.00	0.95256	1	Correct



Intercept	slope	Correct	Incorrect
-11.00	0.10	10	10
-10.50	0.20	12	8
-10.00	0.30	19	1
-8.00	0.20	18	2
-9.00	0.20	15	5



		Actual Value	
		Positive	Negative
Predicated Value	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix

Evaluation Metrix for classification(Confusion matrix)

- 1) Accuracy
- 2) Precision
- 3) Recall
- 4) F-Beta Square

Confusion matrix :

The confusion matrix can be explained by studying the graph below.

		Actual Value	
		Positive	Negative
Predicated Value	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix

Accuracy:

The accuracy function will predict the overall performance of the model.

		Actual Value	
		Positive	Negative
Predicated Value	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$



$$\text{Accuracy} = \frac{\text{Total Correctly Predicted Value}}{\text{Total Value}}$$

Precision

We focus on Precision when It is important to evaluate and **minimize “false positives”**, especially in sensitive domains where false negatives are less of a concern compared to false positives.

High Precision means that the model is effective where we want to minimize **“false positives”**.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad \Rightarrow \quad \text{Precision} = \frac{\text{True Positive}}{\text{Total Predicted Positive}}$$

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Example :
Face recognition:

Recall:

The recall is particularly useful in situations where it is important to **minimize “false negatives”**. For example, in medical diagnostics, it is crucial to identify as many true positive cases as possible, even if it results in a higher number of false positives. In such cases, recall serves as a valuable metric to evaluate the model's ability to identify positive instances.

High recall means that the model is effective when we want to minimize **“false negatives”**.

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad \Rightarrow \quad \text{Recall} = \frac{\text{True Positive}}{\text{Total Actual Positive}}$$

F-1measure:

If both True positive and False Negative is important.

le precision and recall are important

$$F_{\beta} = (1 + \beta^2) \cdot \frac{precision \cdot recall}{(\beta^2 \cdot precision) + recall}$$